

DSA Day 31: Graphs - terminology & spanning trees

Book: Data Structures Through C in Depth - Ch7 §7.1-7.8 · pp 326-334 | Code in Python

Overview

A graph is vertices + edges, directed or undirected. Connectivity, cycles and spanning trees are core concepts.

Key concepts to master

- degree, path, cycle
- connected / strongly connected
- spanning tree vs MST

Python implementation

```
# represent as dict of neighbours  
g={0:[1,2],1:[2],2:[0,3],3:[]}
```

Complexity

Depends on representation.

Common pitfalls

- Mixing directed vs undirected degree
- Assuming a graph is connected

Practice

- Classify 3 graphs
- Define spanning tree vs MST

